Change of variables

(Cylindrical and Spherical coordinates)

1) Triple integrals

Transformation $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

domain $\rightarrow (u,v,w) \longmapsto (x,y,z)$ ← image

where T has continuous first partial derivatives.

Initial expression: $\iiint_D f(x,y,z)\,dx\,dy\,dz$

Change of variables $\Rightarrow \iiint_{D'} f(x(u,v,w), y(u,v,w), z(u,v,w))\,|J_T|\,du\,dv\,dw$

$x = g(u,v,w)$
$y = h(u,v,w)$
$z = k(u,v,w)$

- $J_T = \dfrac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}$ (rows ↓, columns →)

$$|J_T| = \det\left(\frac{\partial(x,y,z)}{\partial(u,v,w)}\right) \quad \text{(determinant)}$$

Theorem: If f is continuous ($D \subseteq \mathbb{R}^3$), then

$$\iiint_D f(x,y,z)\,dx\,dy\,dz = \iiint_{D'} f(x(u,v,w), y(u,v,w), z(u,v,w))\,|J_T|\,du\,dv\,dw$$

(x is a function of u, v and w.)

a) Cylindrical coordinates

$T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$(r,\theta,z) \longmapsto (x,y,z)$

$$\begin{cases} x = r\cos(\theta) \\ y = r\sin(\theta) \\ z = z \end{cases} \quad (*)$$

$* \quad x = g(r,\theta,z) = r\cos(\theta)$
$y = h(r,\theta,z) = r\sin(\theta)$
$z = k(r,\theta,z) = z$

$$J_T = \begin{pmatrix} \cos(\theta) & -r\sin(\theta) & 0 \\ \sin(\theta) & r\cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$ → similar to polar coordinates

$$|J_T| = r$$

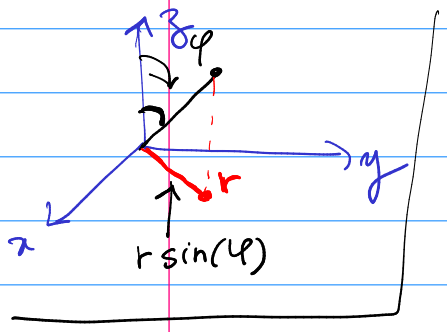
(b) Spherical coordinates

$$T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

cylindrical $r \neq \rho$ $(\rho, \theta, \varphi) \longmapsto (x, y, z)$ with

$$x = \rho\cos(\theta)\sin(\varphi)$$
$$y = \rho\sin(\theta)\sin(\varphi)$$
$$z = \rho\cos(\varphi)$$

$r = \rho\sin(\varphi)$



Exercise : Give the Jacobian matrix of T.

Show this → . $|J_T| = |-\rho^2\sin(\varphi)| = \rho^2\sin(\varphi)$.

3) Examples

1- Consider the volume D that is below the plane $z = 4$, above $z = 1 - (x^2+y^2)$ and within the cylinder $x^2 + y^2 = 1$.

$$\iiint_D K\sqrt{x^2+y^2}\, dV$$

$f(x,y,z) = K\sqrt{x^2+y^2}$ density function

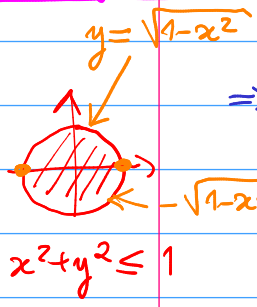
Philosophically

$$\text{Mass} = \int \underbrace{d\text{Mass}}_{\text{density} \,=\, \text{elementary mass}}$$

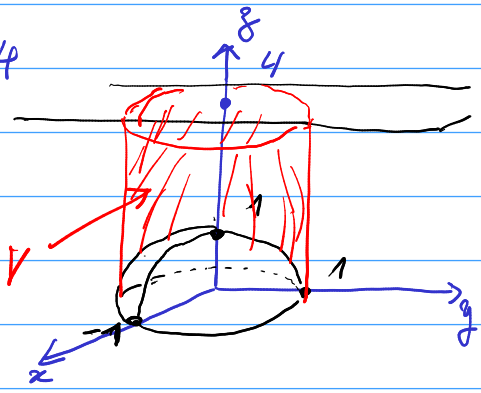
• Cylinder $\begin{cases} x^2+y^2=1 \\ z \text{ free} \end{cases}$ $x^2+y^2 \le 1$

$x^2+y^2=1$

• $\Rightarrow 1-(x^2+y^2) \le z \le 4$



$$\Rightarrow \begin{cases} 1-(x^2+y^2) \le z \le 4 \\ -\sqrt{1-x^2} \le y \le \sqrt{1-x^2} \\ -1 \le x \le 1 \end{cases}$$



$\Rightarrow$ Use cylindrical coordinates :

$$\begin{cases} x = r\cos(\theta) \\ y = r\sin(\theta) \\ z = z \end{cases}$$

• $\theta \in [0; 2\pi]$

• $r \in [0; 1]$

$0 \le r^2 \le x^2+y^2 \le 1$

$0 \le r \le 1$

$1-(x^2+y^2) = 1-r^2$

• $1-r^2 \le z \le 4$

$$\Rightarrow M = \iiint_D k\sqrt{x^2+y^2}\,dx\,dy\,dz = \int_0^{2\pi}\int_0^1\int_{1-r^2}^4 k\sqrt{r^2}\,r\,dz\,dr\,d\theta$$

$\sqrt{r^2} = |r| = r$

$$= k\int_0^{2\pi}\left(\int_0^1\left(\int_{1-r^2}^4 r^2\,dz\right)dr\right)d\theta$$

2- Evaluate $\iiint_D e^{(x^2+y^2+z^2)^{3/2}}\,dx\,dy\,dz$

where D is the sphere $x^2+y^2+z^2 \le 1$.

• $\begin{cases} x = \ell\sin(\varphi)\cos(\theta) \\ y = \ell\sin(\varphi)\sin(\theta) \\ z = \ell\cos(\varphi) \end{cases}$

• $\ell^2 = x^2+y^2+z^2 \le 1$

$\Rightarrow 0 \le \ell \le 1$

$\Rightarrow$ θ and φ are free: $0 \le \theta \le 2\pi$

$0 \le \varphi \le \pi$

$$\Rightarrow \iiint_D e^{(x^2+y^2+z^2)^{3/2}} dx dy dz = \int_0^{\pi}\int_0^{2\pi}\int_0^1 e^{(\rho^2)^{3/2}} \rho^2 \sin(\varphi)\, d\rho d\theta d\varphi$$

$$= \int_0^{\pi} \sin(\varphi) d\varphi \times \int_0^{2\pi} d\theta \times \int_0^1 \rho^2 e^{\rho^3} d\rho.$$

Exercise : Use integrals to evaluate :

1 - Volume of the cylinder $x^2+y^2 = \frac{9}{4}$

$-5 \le z \le 5$.

2 - Volume of the sphere $x^2+y^2+z^2 = 25$.

- Cartesian coordinates

$x^2+y^2+z^2 = 1$

$* \Rightarrow z = \pm\sqrt{1-(x^2+y^2)}$

$\Rightarrow -\sqrt{1-(x^2+y^2)} \le z \le \sqrt{1-(x^2+y^2)}$

$* \sqrt{1-(x^2+y^2)} \Rightarrow 1-(x^2+y^2) \ge 0$

$\Rightarrow x^2+y^2 \le 1$

$\Rightarrow -\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$

$* \sqrt{1-x^2} \Rightarrow 1-x^2 \ge 0$

$\Rightarrow x^2 \le 1$

$\Rightarrow |x| \le 1 \iff -1 \le x \le 1$

$$\Rightarrow \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\left(\int_{-\sqrt{1-(x^2+y^2)}}^{\sqrt{1-(x^2+y^2)}} e^{(x^2+y^2+z^2)^{3/2}} dz\right) dy\, dx$$